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EVALUATING THE DIAGNOSTIC ACCURACY OF SHEAR WAVE ELASTOGRAPHY GUIDED BIOPSY IN BIRADS 4 AND 5 BREAST LESIONS

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BACKGROUND

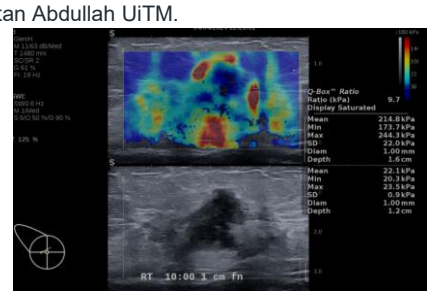
Ultrasound-guided core needle biopsy is a standard method for sampling breast lesions. Despite its high diagnostic accuracy, sampling error may occur due to intratumoral heterogeneity. Shear wave elastography (SWE) may identify areas of differing tissue stiffness and potentially guide targeted sampling. This study evaluates SWE characteristics between high- and low-stiffness areas within the same breast lesion and their association with histopathological diagnosis.

OBJECTIVES

To evaluate the differences in SWE characteristics between high- and low-stiffness areas within the same BI-RADS 4 and 5 breast lesions and assess their association with histopathological diagnosis.

METHODS

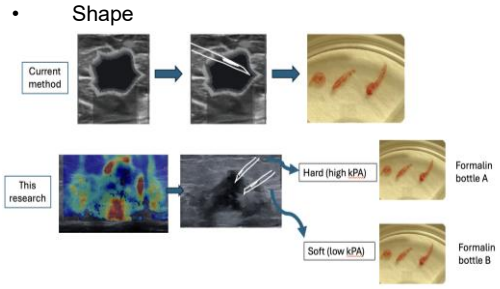
This cross-sectional diagnostic study included patients with BI-RADS 4 and 5 breast lesions referred for ultrasound-guided core needle biopsy and SWE assessment in Radiology Department Hospital Al-Sultan Abdullah UiTM.



Quantitative SWE parameters and Qualitative SWE color scores were recorded and correlated with histopathological findings

Quantitative
- Minimum stiffness values, E^{min}
- Maximum stiffness values, E^{max}
- Mean stiffness values, E^{mean}
- Ratio between the mean stiffness in the mass and the fatty tissue

Qualitative
- Color score
- Homogeneity
- Shape



The Wilcoxon signed-rank test was used for paired comparisons between hard and soft regions, while the Mann–Whitney U test compared SWE parameters between benign and malignant lesions. A p-value <0.05 was considered statistically significant.

RESULTS AND ANALYSIS

Total of 37 samples acquired with SWE parameters from 33 patients.
Among the tumors, 70% (n:26) are benign and 30% (n:11) are malignant.

Demographic data of the 37 samples

Variables	Benign, n:26 (70%)	Malignant, n: 11 (30%)
Age	36.9 ± 16.3 years (range 16 to 68 years)	53.5 ± 13.2 years (range 30 to 75 years)
Size (mm)	25.8 ±13.0 (range 1 to 55 mm)	23.6± 5.6 (range 17 to 35mm)
BI-RADS, n (%) 4a 4b 4c 5	 18, 69% 5, 19 % 0, 0% 3, 11%	 2, 18% 1, 9 % 1, 9% 7, 64 %

Among 37 lesions with complete quantitative SWE data, maximum and mean SWE values were significantly higher in hard than soft regions (both p<0.001), with higher values observed in all 37 lesions. The hard-region SWE ratio was also significantly higher than the soft-region ratio (p<0.001).

SWE stiffness within the same lesion

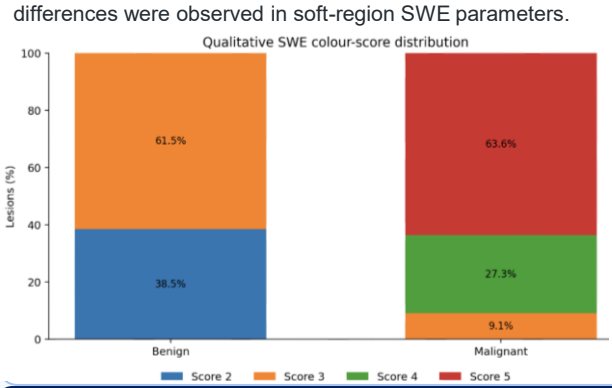
Hard region
Maximum SWE ↑
Mean SWE ↑
High stiffness

≠
p < 0.001

Soft region
Minimum SWE ↓
Mean SWE ↓
Lower stiffness

Hard regions had significantly higher maximum and mean SWE values than soft regions (both p < 0.001).

Among benign (n=26) and malignant (n=11) lesions, malignant lesions demonstrated higher maximum and mean hard-region SWE values, although these differences did not reach statistical significance (p=0.065 and p=0.058, respectively). No significant differences were observed in soft-region SWE parameters.



Qualitative SWE demonstrated a distinct shift toward higher colour scores in malignant lesions. Among malignant lesions, 63.6% had a colour score of 5 and 27.3% had a score of 4, whereas all benign lesions had colour scores of 2 or 3.

CONCLUSION

SWE demonstrated significant stiffness heterogeneity within breast lesions. While quantitative SWE did not significantly differentiate benign from malignant lesions, higher qualitative SWE colour scores were associated with malignancy. SWE may therefore complement conventional ultrasound in identifying areas for targeted biopsy.

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